

INK-Merge

Merge with the functional corrections the current model still needs

1 Models & signals

Unlabeled calibration
 $x \in \mathcal{C}_t$

Frozen experts
 $\{\theta_t\}_{t=1}^T$
cached predictions

Current merge
 $\theta^{(r)}$
forward pass

2 Residual-driven geometry

From prediction mismatch to a layerwise correction

$e_{t,c}^{(r)} = p_{t,c} - q_c^{(r)}$

Signed class residuals

$u_t^{(r)} = -\frac{\nabla_{h_\ell} D_{\text{KL}}(p_t \| q^{(r)})}{d_t^{(r)}}$

$d_t^{(r)} = \sqrt{\sum_c \frac{(e_{t,c}^{(r)})^2}{q_c^{(r)}} + \varepsilon}$

Pearson normalization · stop gradient through d

Input geometry

$A_t^{(r)} = \mathbb{E}[hh^\top]$
 $h = h_{\ell-1}$

Correction geometry

$G_t^{(r)} = \mathbb{E}[u_t u_t^\top]$
 $p_t = q^{(r)} \Rightarrow u_t = 0$

3 Merge & validate

Two-sided layerwise solve

$$\sum_t A_t^{(r)} X^\star G_t^{(r)}$$

$$= \sum_t A_t^{(r)} X_t G_t^{(r)}$$

Matrix-free preconditioned CG

Held-out teacher-KL line search

$$W_b^{(r+1)} = W_b^{(r)} + \omega_b (W_b^\star - W_b^{(r)})$$

$$\omega_b \in \Omega \subseteq [0, 1], \quad 0 \in \Omega$$

Disjoint $\mathcal{H}_t$ · zero step can reject an update

Update $q^{(r+1)}$ and re-estimate both A and G

Best held-out round $\hat{\theta}$